

Flow Dataset Cleaning Report

How the 10 raw CICIDS2018 traffic files were merged and cleaned into one machine-learning-ready dataset — every step, every row removed, and the final structure.

Raw input 16,233,002 rows · ~6.9 GB	Cleaned dataset 11,872,582 rows · ~4.8 GB	Columns 79 54 int · 24 float · 1 text	Attack classes 15 incl. Benign	Validation 26 / 0 checks passed / failed
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01 — SOURCE DATA

Ten daily capture files from CIC-IDS-2018

Each file is one day of network traffic recorded by CICFlowMeter, which turns raw packets into one row per network flow with 80 statistical features and a Label. Captured by the Canadian Institute for Cybersecurity, Feb–Mar 2018.

Capture day	Rows	Attack classes present
Wed 14-02	1,048,575	Benign, FTP-BruteForce, SSH-Bruteforce
Thu 15-02	1,048,575	Benign, DoS-GoldenEye, DoS-Slowloris
Fri 16-02	1,048,575	Benign, DoS-Hulk, DoS-SlowHTTPTest
Tue 20-02	7,948,748	Benign, DDoS-LOIC-HTTP (4 extra ID columns)
Wed 21-02	1,048,575	Benign, DDoS-HOIC, DDoS-LOIC-UDP
Thu 22-02	1,048,575	Benign, Brute-Force Web/XSS, SQL Injection
Fri 23-02	1,048,575	Benign, Brute-Force Web/XSS, SQL Injection
Wed 28-02	613,071	Benign, Infiltration
Thu 01-03	331,100	Benign, Infiltration
Fri 02-03	1,048,575	Benign, Bot
Merged	16,233,002	15 classes total

The Tuesday file carried four extra identifier columns (Flow ID, Src IP, Src Port, Dst IP) the other nine lacked; these were dropped at merge so every file shares the same 80 columns.

02 — CLEANING PIPELINE

Ten ordered steps, streamed in chunks to stay within 12 GB of RAM

#	Step	Rows in	Rows out	Dropped
1	Load & merge (remove 59 stray header rows)	16,233,002	16,232,943	59
2	Drop Timestamp column (80 → 79 cols)	16,232,943	16,232,943	0
3	Remove inf / NaN / null values	16,232,943	16,137,183	95,760
4	Remove corrupt rows (neg / >120 s timings)	16,137,183	16,137,168	15

#	Step	Rows in	Rows out	Dropped
5	Remove exact duplicate rows	16,137,168	11,916,123	4,221,045
6	Remove conflicting labels	11,916,123	11,872,582	43,541
7	Fix data types (int / float / text)	11,872,582	11,872,582	0
8	Shuffle rows (seed 42)	11,872,582	11,872,582	0
9	Save final CSV	11,872,582	11,872,582	0
10	Validate (26 checks passed, 0 failed)	11,872,582	11,872,582	0
Net		16,233,002	11,872,582	4,360,420

Net result: 4,360,420 rows removed (26.9%). The largest cut was 4.22M exact duplicate flows — without the Timestamp column, repeated attack connections become byte-for-byte identical rows.

Each step reads the previous step's file, removes one class of problem, and records its rows in and out. The rules: corrupt = any negative value (window size may be -1) or a timing column above the 120-second flow timeout; conflicting labels = identical feature rows carrying more than one Label. All exact duplicates were removed by choice, even though that left FTP-BruteForce and SlowHTTPTest with only a few dozen rows.

03 — FINAL COLUMNS & TYPES

All 79 columns, in file order

Type colour: **int64** (54) **float64** (24) **str** (1) * = all-zero column (kept for completeness)

1 Dst Port int64	41 Pkt Len Max int64
2 Protocol int64	42 Pkt Len Mean float64
3 Flow Duration int64	43 Pkt Len Std float64
4 Tot Fwd Pkts int64	44 Pkt Len Var float64
5 Tot Bwd Pkts int64	45 FIN Flag Cnt int64
6 TotLen Fwd Pkts int64	46 SYN Flag Cnt int64
7 TotLen Bwd Pkts int64	47 RST Flag Cnt int64
8 Fwd Pkt Len Max int64	48 PSH Flag Cnt int64
9 Fwd Pkt Len Min int64	49 ACK Flag Cnt int64
10 Fwd Pkt Len Mean float64	50 URG Flag Cnt int64
11 Fwd Pkt Len Std float64	51 CWE Flag Count int64
12 Bwd Pkt Len Max int64	52 ECE Flag Cnt int64
13 Bwd Pkt Len Min int64	53 Down/Up Ratio int64
14 Bwd Pkt Len Mean float64	54 Pkt Size Avg float64
15 Bwd Pkt Len Std float64	55 Fwd Seg Size Avg float64
16 Flow Byts/s float64	56 Bwd Seg Size Avg float64
17 Flow Pkts/s float64	57 Fwd Byts/b Avg * int64
18 Flow IAT Mean float64	58 Fwd Pkts/b Avg * int64
19 Flow IAT Std float64	59 Fwd Blk Rate Avg * int64
20 Flow IAT Max int64	60 Bwd Byts/b Avg * int64
21 Flow IAT Min int64	61 Bwd Pkts/b Avg * int64
22 Fwd IAT Tot int64	62 Bwd Blk Rate Avg * int64

23	Fwd IAT Mean	float64	63	Subflow Fwd Pkts	int64
24	Fwd IAT Std	float64	64	Subflow Fwd Byts	int64
25	Fwd IAT Max	int64	65	Subflow Bwd Pkts	int64
26	Fwd IAT Min	int64	66	Subflow Bwd Byts	int64
27	Bwd IAT Tot	int64	67	Init Fwd Win Byts	int64
28	Bwd IAT Mean	float64	68	Init Bwd Win Byts	int64
29	Bwd IAT Std	float64	69	Fwd Act Data Pkts	int64
30	Bwd IAT Max	int64	70	Fwd Seg Size Min	int64
31	Bwd IAT Min	int64	71	Active Mean	float64
32	Fwd PSH Flags	int64	72	Active Std	float64
33	Bwd PSH Flags *	int64	73	Active Max	int64
34	Fwd URG Flags	int64	74	Active Min	int64
35	Bwd URG Flags *	int64	75	Idle Mean	float64
36	Fwd Header Len	int64	76	Idle Std	float64
37	Bwd Header Len	int64	77	Idle Max	int64
38	Fwd Pkts/s	float64	78	Idle Min	int64
39	Bwd Pkts/s	float64	79	Label	str
40	Pkt Len Min	int64			

04 — CLASS DISTRIBUTION

Heavily imbalanced — Benign is 88.8% of all rows

Class	Rows	Share
Benign	10,543,008	88.80%
DDoS attacks-LOIC-HTTP	575,364	4.85%
DDOS attack-HOIC	198,861	1.67%
DoS attacks-Hulk	145,199	1.22%
Bot	144,535	1.22%
Infiltration	117,598	0.99%
SSH-Bruteforce	94,041	0.79%
DoS attacks-GoldenEye	41,406	0.35%
DoS attacks-Slowloris	9,908	0.08%
DDOS attack-LOIC-UDP	1,730	0.01%
Brute Force -Web	547	0.0046%
Brute Force -XSS	226	0.0019%
SQL Injection	77	0.0006%
DoS attacks-SlowHTTPTest	43	0.0004%
FTP-BruteForce	39	0.0003%

The imbalance and the two tiny classes (FTP-BruteForce 39, SlowHTTPTest 43) are traits to handle at model time — with a stratified split and class weighting — not defects in the cleaned data.

05 — TOOLS & TECHNIQUES

How the cleaning was built

Libraries: Python 3.13 · pandas 3.0 · numpy 2.5 · pyarrow 25 (Parquet) · scikit-learn 1.9

Techniques: chunked Parquet streaming · single row-group reads · 64-bit row hashing for duplicate detection · feature-vector hashing for label conflicts · bucketed on-disk shuffle (seed 42)

The full dataset never fits in memory, so every step streams 100k-row chunks and keeps only small running totals. The final CSV passed an independent validator that re-reads it from disk: no NaN / null / inf, no duplicates, no conflicting labels, one data type per column, all values within float32 range, and rows shuffled — 26 checks passed, 0 failed.

Figures generated from `cleaning_log.json` and `9_final_summary.json`. Report compiled 2026-09-28.